

MILESTONE 4: Frontend Development – LegalEase

Milestone 4 focuses on creating a responsive and user-centric web interface for LegalEase using Streamlit. This milestone enables users to easily input legal details, generate AI-powered documents, preview the results, make inline edits, and download documents in multiple formats (TXT, DOCX, PDF) with rich formatting and embedded branding.

Activity 4.1: Designing and Developing the User Interface

Step 1: Streamlit Page Configuration and Layout Setup

Configured Streamlit app with:

Added a three-column layout using `st.columns` to center-align the company logo

Step 2: Header and Title

Displayed a subtle center-aligned title under the logo:

Step 3: User Input Interface

Created intuitive inputs for:

- o `document_type`: type of legal document

- o parties: involved stakeholders

- o terms: semicolon-separated clauses

- o dates: effective date

Activity 4.2: Creating Dynamic Previews and Export Options

Step 1: Generate Document with Backend AI

On clicking "Generate Document", a POST request is sent to the FastAPI backend:

AI-generated content is sanitized and displayed dynamically:

Step 2: HTML Preview Rendering

The output is previewed in an elegant dark-themed scrollable card with semantic

HTML:

Step 3: Editable Document Preview

The user can click "Edit Document" and modify the generated content using a

text area:

Step 4: Multi-Format Download Options

Users can download the document as:

- o `.txt` (Plain Text)

- o `.docx` (Formatted Word Document with logo, headings, and table)

- o `.pdf` (Branded PDF with logo and footer)

Core Functions & Utilities

`sanitize_text(text)`

Removes special characters and typographic quotes to ensure clean formatting.

`format_docx(text, doc_type)`

Uses `python-docx` to:

- o Embed logo

- o Add titles, paragraph formatting

- o Auto-generate a Terms table from semicolon-separated input

`format_pdf(text, doc_type)`

Utilizes `FPDF` with custom header/footer:

- o Center-aligned logo

- o Bold headings for sections

- o Bullet-style terms

`format_html_preview(text)`

Converts output to stylized HTML blocks for inline display.